

Project Design Phase-II

Data Flow Diagram & User Stories

Date	19 July 2026
Team ID	XXXX XXXX XXXX
Project Name	Ucab – Smart Cab Booking Application
Maximum Marks	4 Marks

Data Flow Diagram (Level-0 Description)

User (Browser / React Client) → REST API (Express.js) → Business Logic (Node.js) → Database (MongoDB).

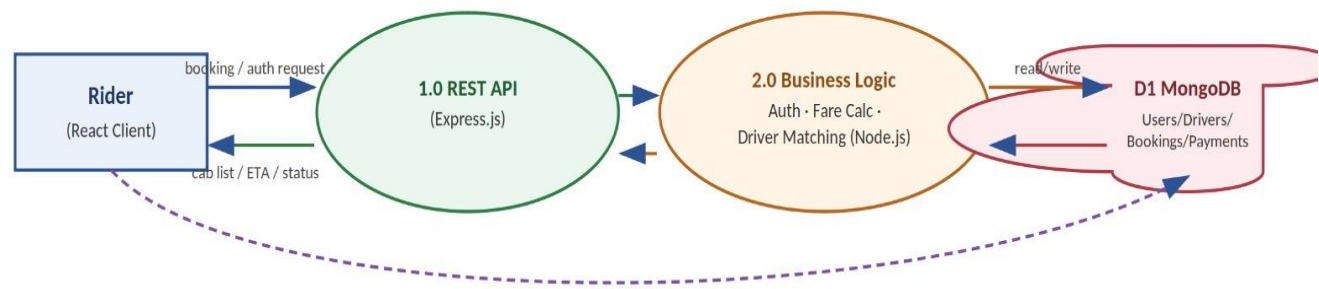
The client sends booking, auth, and payment requests to the Express API; Node.js processes business rules (fare calculation, driver matching); MongoDB stores users, drivers, bookings, and payment records; responses (cab list, ETA, trip status) flow back to the client, with live location updates pushed via WebSockets/Socket.io.

User Stories

User Type	Epic	User Story No.	User Story / Task	Acceptance Criteria	Priority	Release
Rider	Registration	USN-1	As a user, I can register with email, password & confirm password	Account created & login enabled	High	Sprint-1
Rider	Registration	USN-2	As a user, I receive a confirmation email after registering	Confirmation email received	High	Sprint-1
Rider	Login	USN-3	As a user, I can log in with email & password	Redirected to dashboard	High	Sprint-1
Rider	Booking	USN-4	As a user, I can enter pickup & drop-off locations on a map	Locations saved to booking	High	Sprint-2

User Type	Epic	User Story No.	User Story / Task	Acceptance Criteria	Priority	Release
Rider	Booking	USN-5	As a user, I can view nearby cabs with fare & ETA before booking	Cab list with fare/ETA shown	High	Sprint-2
Rider	Tracking	USN-6	As a user, I can track my assigned cab live on the map	Live marker updates on map	High	Sprint-3
Rider	Payment	USN-7	As a user, I can pay automatically via a saved payment method	Payment auto-charged post-ride	High	Sprint-3
Rider	History	USN-8	As a user, I can view my past bookings and receipts	History list with details	Medium	Sprint-4
Rider	Add-ons	USN-9	As a user, I can apply a discount code at checkout	Discount reflected in fare	Low	Sprint-4
Admin	Management	USN-10	As an admin, I can view/manage all users, drivers and bookings	Admin dashboard access	Medium	Sprint-4

Data Flow Diagram (Level-0) — Ucab



Real-time channel (Socket.io): live driver location → rider client

Driver client pushes location updates into the same REST/WebSocket layer, broadcast back to the rider client.